

The Magnetic Moment of the Electron*

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The spin magnetic dipole moment of the electron is computed to fourth order in perturbation theory. The procedure is to consider a system of one electron moving in a constant external magnetic field. The Green's function and mass operator for the electron are computed as functions of this field. The expectation value of the mass operator is the self-energy of the electron and the magnetic moment is identified from that part of the self-energy linear in the external field. The techniques used are illustrated in a calculation of the second-order moment. The derivation of the fourth-order moment is sketched. It is found to be $\mu^{(4)} = -0.328 (\alpha/\pi)^2$ in Bohr magnetons, where α is the fine-structure constant. This differs from the result of a previous calculation.

1. INTRODUCTION

In a previous note (1), we reported the results of a new calculation of the anomalous magnetic moment of the electron to fourth order in perturbation theory, which result differed from the earlier value obtained by Karplus and Kroll (2). In addition, we discussed the effects of the new moment on various aspects of atomic physics. It is the object of the present paper to exhibit the calculation itself, in some detail.

The theory of the spin magnetic dipole moment of the electron had its beginnings in 1928, when, on the basis of his equation for a relativistic electron interacting with an external magnetic field $\mathbf{H}$, Dirac (3) found that the energy of the electron contained a term $-(e\hbar/2mc)\boldsymbol{\sigma}\cdot\mathbf{H}$. (The components of $\boldsymbol{\sigma}$ are the three Pauli spin matrices). The identification of $\mu = e\hbar/2mc$ as the magnetic moment followed immediately.

The subsequent quantization of the electromagnetic and electron fields required that this be corrected to take into account the possibility of radiation of electromagnetic quanta by the electron and polarization of the electron vacuum by the electromagnetic field with electron-positron pairs. These so-called radi-

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ative corrections had to be treated by perturbation theory. The second-order correction of Schwinger (4) and the fourth-order correction of Karplus and Kroll changed Dirac's moment to

$$\mu = (e\hbar/2mc)[1 + \frac{1}{2}(\alpha/\pi) - 2.973(\alpha/\pi)^2].$$

Karplus and Kroll used the techniques of Feynman and Dyson for constructing matrix elements of observable quantities. As pioneers in the hitherto unexplored fourth order, they were interested in seeing if these techniques remained valid and if the renormalization program could be carried through in a consistent way. Their results indicated that everything behaved as expected. The major difficulty was the length of the calculation. To provide an independent procedure and in an attempt to simplify the structure of the calculation, we here use the mass-operator formalism of Schwinger (5). The mass operator may be thought of as describing the mass or self-energy of an electron in interaction with the electromagnetic field. If we impose a constant (in space and time) external magnetic field on the system of one electron, the expectation value of the mass operator, computed to the desired order of perturbation theory, will contain a term which is the self-energy of the electron in such a field. The magnetic moment is identified from the coefficient of that term in the energy which is linear in the magnetic field.

In Section 2, after deriving the normal Dirac moment, we define the mass and polarization operators in terms of the differential equations for the Green's functions of the electron and photon fields. The usual procedure is to define these Green's functions in the limit of a vanishing external electromagnetic field. The presence of this external field is then taken into account in the same way as the radiative corrections due to the virtual photon field. On the other hand, our Green's functions are defined to describe the propagation of electrons and photons in the full presence of the external field we have imposed. The inclusion of effects due to the field¹ is accomplished by replacing the operator $(\hbar/i)(\partial/\partial x_\mu)$ for the mechanical momentum of the electron by $(\hbar/i)(\partial/\partial x_\mu) - (e/c)A_\mu(x)$ when the electron moves in a field characterized by the potential $A_\mu(x)$. With the definition of a vertex operator the theory is cast in a form amenable to perturbation calculations. There follows a discussion on the extraction of the field-dependent self-energy from the mass operator. The renormalization of the various operators is carried out so that they correspond to physically observable quantities. Finally, by a series of iterations on the unperturbed and second-order operators, an expression is generated for the mass operator in fourth-order perturbation theory.

In order to illustrate the method of calculation, we spend some time in Section

¹ When we say "field" without any qualifying adjectives, we mean the external electromagnetic field.

3 deriving the second-order moment. This involves an integration over the possible four-momenta and a sum over the possible polarizations of a virtual intermediate photon. A method for carrying out the integration is described in detail since it is also used in the fourth-order calculation. If we ignore terms which are of higher degree than first in the external field, the integral turns out to be only slightly more complicated than it would have been in the absence of this field.

Section 4 is devoted to a description of the second-order Green's functions and vertex operator and their contributions to the fourth-order mass operator. The resultant effects on the self-energy are invariably in the form of complicated integrals over auxiliary variables which had been introduced in order to simplify the momentum-space integrals. Changes of these variables are specified which serve to bring the integrals into manageable forms. The total fourth-order moment is presented in closed form.

In an Appendix we present another method of deriving the moment—which has the advantage that the divergences are eliminated at the beginning, but which seems to involve more work.

2. THE MASS-OPERATOR FORMALISM

The Dirac equation for an electron moving in the presence of an external electromagnetic potential $A_\mu(x)$ is

$$\{\gamma_\mu[p_\mu - eA_\mu(x)] + m\}\psi_A(x) = 0. \quad (2.1)$$

The notation is as follows:

$$\hbar = c = 1;$$

$$x_\mu = (\mathbf{x}, ix_0) = (\mathbf{x}, it), (\mu = 1, 2, 3, 4);$$

$$a_\mu b_\mu = \mathbf{a} \cdot \mathbf{b} - a_0 b_0 = ab;$$

$$p_\mu = -i\partial/\partial x_\mu;$$

$$\{\gamma_\mu, \gamma_\nu\} = \gamma_\mu \gamma_\nu + \gamma_\nu \gamma_\mu = -2\delta_{\mu\nu}, \gamma_\mu^\dagger = -\gamma_\mu, \gamma_4 = i\gamma_0 = i\beta;$$

$$\bar{\psi}(x) = \psi^\dagger(x)\beta.$$

The mass and charge of the electron are represented by m and e , respectively.

We shall demonstrate that the magnetic moment may be thought of as contributing to a field-dependent mass or self-energy of the electron. For simplicity we impose a constant (in space and time) external field, $F_{\mu\nu}$, given by

$$F_{\mu\nu} = \partial A_\nu/\partial x_\mu - \partial A_\mu/\partial x_\nu.$$

In particular, if we take $A_\mu = -\frac{1}{2}F_{\mu\nu}x_\nu$, Eq. (1) reads

$$(\gamma p + \frac{1}{2}e\gamma Fx + m)\psi_A(x) = 0.$$

Interpreting $\frac{1}{2}e\gamma Fx$ as an operator which represents an addition to the mass of the electron due to the action of F , we may evaluate the effect to first order in F by taking its expectation value in a state $|\psi_0\rangle$ of the free electron whose wave function, $\langle x | \psi_0 \rangle = \psi_0(x)$, satisfies $(\gamma p + m)\psi_0(x) = 0$. That is,

$$\begin{aligned} (\Delta_F m)^{(0)} &= \langle \bar{\psi}_0 | \frac{1}{2}e\gamma Fx | \psi_0 \rangle = -(e/4m)\langle \bar{\psi}_0 | (\gamma p)(\gamma Fx) + (\gamma Fx)(\gamma p) | \psi_0 \rangle \\ &= -(e/4m)\langle \bar{\psi}_0 | -i\gamma_\mu\gamma_\nu F_{\nu\lambda}\delta_{\mu\lambda} - 2\delta_{\mu\nu}F_{\nu\lambda}x_\lambda p_\mu | \psi_0 \rangle \\ &= -(e/2m)\langle \bar{\psi}_0 | (\frac{1}{2}\sigma_{\mu\nu}F_{\mu\nu}) + (\frac{1}{2}L_{\mu\nu}F_{\mu\nu}) | \psi_0 \rangle, \end{aligned}$$

where $L_{\mu\nu} = x_\mu p_\nu - x_\nu p_\mu$ and $\sigma_{\mu\nu} = \frac{1}{2}i[\gamma_\mu, \gamma_\nu]$. This is just the change in mass to be expected of a particle with spin $\frac{1}{2}$ and relativistic orbital angular momentum $L_{\mu\nu}$, and hence we say that the spin magnetic moment of the electron is 1 Bohr magneton.

We want to find the additional field-dependent change in the self-energy of the electron, in interaction not only with a constant external field, but also with the virtual fluctuations in the electromagnetic field. Schwinger (5) has shown that the Dirac equation may then be modified to read

$$\gamma[p - e_0 A(x)]\psi_A(x) + \int (dx')M(x, x')\psi_A(x') = 0, \quad (2.2)$$

where $M(x, x')$ is called the mass function and where $\psi_A(x)$ is still a numerical wave function. The parameter e_0 represents the "bare charge". The corresponding causal one-electron Green's function satisfies

$$\gamma[p - e_0 A(x)]G(x, x') + \int (dx'')M(x, x'')G(x'', x') = \delta(x - x'). \quad (2.3)$$

In the Lorentz gauge the causal one-photon Green's function equation is

$$-(\partial^2/\partial\xi_\lambda^2)\mathcal{G}_{\mu\nu}(\xi, \xi') + \int (d\xi'')P_{\mu\lambda}(\xi, \xi'')\mathcal{G}_{\lambda\nu}(\xi'', \xi') = \delta_{\mu\nu}\delta(\xi - \xi'), \quad (2.4)$$

where $P_{\mu\nu}(\xi, \xi')$ is called the polarization function. It describes the interaction of photons with virtual electron-positron pairs.

We shall distinguish between the electron and photon coordinates by using x and ξ , respectively; in momentum space, p and k . It is convenient (6) to think of a function $O(x, x')$ as the matrix element of an operator O : $\langle x | O | x' \rangle$. The matrix elements of O in momentum space may be obtained from the Fourier transformation

$$O(p, p') = (2\pi)^{-4} \int (dx)(dx')e^{-ipx}O(x, x')e^{ip'x'},$$

which is equivalent to the matrix transformation

$$\langle p | O | p' \rangle = \int \langle p | x \rangle \langle dx \rangle \langle x | O | x' \rangle \langle dx' \rangle \langle x' | p' \rangle$$

with $\langle x | p \rangle = (2\pi)^{-2} e^{ipx}$. Some special matrices are

$$\begin{aligned} \langle x | 1 | x' \rangle &= \delta(x - x'), \\ \langle x | p_\mu | x' \rangle &= -i(\partial/\partial x_\mu) \delta(x - x'), \\ \langle x | A_\mu | x' \rangle &= A_\mu(x) \delta(x - x'), \\ \langle x | \gamma_\mu(\xi) | x' \rangle &= \gamma_\mu \delta(x - x') \delta(\xi - x), \\ \langle x | \gamma_\mu(k) | x' \rangle &= \gamma_\mu e^{-ikx} \delta(x - x'). \end{aligned} \tag{2.5}$$

If we suppress all indices, Eqs. (3) and (4) assume the abstract forms

$$[\gamma\Pi + M]G = 1, \tag{2.6}$$

$$[k^2 + P]\mathfrak{S} = 1, \tag{2.7}$$

in which

$$\Pi = p - e_0 A. \tag{2.8}$$

The introduction of a “vertex” or “interaction” operator

$$\Gamma_\mu(\xi) = -[\delta/\delta e_0 A_\mu(\xi)]G^{-1} = \gamma_\mu(\xi) - [\delta/\delta e_0 A_\mu(\xi)]M \tag{2.9}$$

allows one to write

$$M = m_0 + ie_0^2 \int (d\xi)(d\xi') \gamma_\mu(\xi) G \Gamma_\nu(\xi') \mathfrak{S}_{\nu\mu}(\xi', \xi), \tag{2.10}$$

$$P_{\mu\nu}(\xi, \xi') = -ie_0^2 \text{tr} \int (dx)(dx')(dx'') \gamma_\mu(\xi) G(x, x') \Gamma_\nu(x', x''; \xi') G(x'', x), \tag{2.11}$$

where m_0 is the “bare-mass” parameter and where “tr” indicates a sum over diagonal matrix elements.

The operator M is a relativistic scalar matrix in the same algebra as the γ -matrices and is an operator function of the operator Π . All such scalar operators not explicitly dependent on the constant external field F can be expressed in terms of $\gamma\Pi$. Scalar matrix operators linear in F are $\gamma F\gamma$ and $\gamma F\Pi$, while F^2 , $\gamma F F \Pi$ and $\Pi F F \Pi$ are quadratic in F ; and so on. We write M as follows:²

$$M = m_0 + M_0[\gamma\Pi] + M_F[\gamma\Pi, \mathfrak{F}] \tag{2.12}$$

² Boldface brackets will be used to exhibit functional dependence on operators.

so that

$$G = \{\gamma\Pi + m_0 + M_0[\gamma\Pi] + M_F[\gamma\Pi, \mathfrak{F}]\}^{-1}. \quad (2.13)$$

The symbol $\mathfrak{F}$ represents all scalar operators explicitly dependent on F . We require

$$\lim_{F \rightarrow 0} M_F[\gamma\Pi, \mathfrak{F}] = M_F[\gamma p, 0] = 0.$$

In the absence of an external field, G must possess a pole at $\gamma\Pi = -m$, in order that it represent the physical electron of mass m . We expand $M_0[\gamma\Pi]$ in a Taylor series about $\gamma\Pi = -m$:

$$M_0[\gamma\Pi] = M_0[-m] + (\gamma\Pi + m)\bar{M}_0 + M_1[\gamma\Pi]. \quad (2.14)$$

The number $\bar{M}_0$ is given by

$$\bar{M}_0 = (\partial/\partial m)M_0[\gamma\Pi] |_{\gamma\Pi + m = 0}. \quad (2.15)$$

In the limit $F = 0$, $\gamma\Pi \rightarrow -m$, we require that

$$M_0[-m] = m - m_0,$$

whence

$$\begin{aligned} G[F = 0, \gamma\Pi \rightarrow -m] &\sim \{\gamma p + m + (\gamma p + m)\bar{M}_0\}^{-1} \\ &= (1 + \bar{M}_0)^{-1}(\gamma p + m)^{-1}. \end{aligned} \quad (2.16)$$

The Green's function of the physical electron should behave near the pole like $(\gamma p + m)^{-1}$. Accordingly, we define the renormalized electron Green's function, G' , as

$$\begin{aligned} G' &= (1 + \bar{M}_0)G \\ &= \{\gamma\Pi + m + (1 + \bar{M}_0)^{-1}(M_1[\gamma\Pi] + M_F[\gamma\Pi, \mathfrak{F}])\}^{-1} \\ &= (\gamma\Pi + m + M')^{-1}. \end{aligned} \quad (2.17)$$

We renormalize the photon Green's function in a manner similar to our treatment of the mass operator and electron Green's function. The polarization operator must be a function of k and F only:

$$P = P_0[k] + P_F[k, F] \quad (2.18)$$

such that $P_F[k, F] \rightarrow 0$ as $F \rightarrow 0$. We write

$$P_0[k] = k^2 \bar{P}_0 + P_1[k]. \quad (2.19)$$

The term $P_0[k^2 = 0]$, which would correspond to a physical photon mass, is absent since the principle of gauge invariance guarantees that both the bare

and physical photon masses are zero. Thus we define a renormalized photon Green's function

$$\begin{aligned} \mathcal{G}' &= \mathcal{G}(1 + \bar{P}_0) \\ &= \{k^2 + (1 + \bar{P}_0)^{-1}(P_1[k] + P_r[k, F])\}^{-1} \\ &= (k^2 + P')^{-1} \end{aligned} \quad (2.20)$$

so that $\mathcal{G}' \sim 1/k^2$ for $k^2 \rightarrow 0$ and $F \rightarrow 0$.

Since $\mathcal{G}$ is a vacuum expectation value quadratic in A_μ , we must define a renormalized electromagnetic potential

$$A_{\mu'} = (1 + \bar{P}_0)^{1/2} A_\mu. \quad (2.21)$$

In order to preserve the gauge-invariant structure of Π as it appears in Eq. (17), we identify the physical charge e as

$$e = (1 + \bar{P}_0)^{-1/2} e_0. \quad (2.22)$$

If the external field were made vanishingly small, Eq. (9) could be integrated to yield

$$G^{-1} = \gamma p + m_0 + M_0[\gamma p] - e \int (d\xi) \Gamma_{\mu 0}(\xi) A_{\mu'}'(\xi), \quad (2.23)$$

where $\Gamma_{\mu 0}$ is the field-independent part of Γ_μ , possessing the general structure

$$\Gamma_{\mu 0} = \gamma_\mu h[\gamma p, \gamma k]$$

(in which the γ_μ may appear imbedded among the other factors). Near the pole $\gamma p = -m$, and in the limit of interaction with a very low energy ($k = 0$) photon, we have $\Gamma_{\mu 0} \sim \gamma_\mu h[-m, 0]$ so that

$$G^{-1} \sim (\gamma p + m)(1 + \bar{M}_0) - e \gamma A' h[-m, 0].$$

In order that gauge invariance be maintained, the structure $p - eA'$ must be preserved, which implies

$$h[-m, 0] = 1 + \bar{M}_0.$$

By defining a renormalized interaction operator

$$\Gamma_{\mu'} = \Gamma_\mu (1 + \bar{M}_0)^{-1}, \quad (2.24)$$

we obtain an expression for G'^{-1} corresponding to (23):

$$G'^{-1} = \gamma p + m + M_1[\gamma p] - e \int (d\xi) \Gamma_{\mu'}'(\xi) A_{\mu'}'(\xi). \quad (2.25)$$

We may now express the reduced mass operator, M' , in terms of the renor-

malized quantities e , G' , $\mathfrak{G}'$, and Γ' :

$$M' = ie^2(1 + \bar{M}_0)^{-1} \int_r (d\xi)(d\xi') \gamma_\mu(\xi) G' \Gamma'_\nu(\xi') \mathfrak{G}'_{\nu\mu}(\xi', \xi). \quad (2.26)$$

The subscript r indicates the removal of the first two terms of the expansion in powers of $(\gamma\Pi + m)$ of the F -independent part of the integral. Likewise, the reduced polarization operator, P' , is given by

$$\begin{aligned} P'_{\mu\nu}(\xi, \xi') = & -ie^2(1 + \bar{M}_0)^{-1} \text{tr} \int_r (dx)(dx')(dx'') \gamma_\mu(\xi) \\ & \times G'(x, x') \Gamma'_\nu(x', x''; \xi') G'(x'', x), \end{aligned} \quad (2.27)$$

where the subscript r now indicates the removal of the first two terms of the expansion in powers of k^2 . In momentum space, these formulas are

$$M' = i(\alpha/4\pi^3)(1 + \bar{M}_0)^{-1} \int_r (dk)(dk') \gamma_\mu(-k) G' \Gamma'_\nu(k') \mathfrak{G}'_{\nu\mu}(-k', -k) \quad (2.28)$$

$$\begin{aligned} P' = & -i(\alpha/4\pi^3)(1 + \bar{M}_0)^{-1} \text{tr} \int_r (dx)(dx')(dx'') \gamma_\mu(k) \\ & \times G'(x, x') \Gamma'_\nu(x', x''; -k') G'(x'', x), \end{aligned} \quad (2.29)$$

where $\alpha = e^2/4\pi$ is the fine-structure constant.

We consider the quantities we have defined to various orders in e . To lowest order:

$$\begin{aligned} M'^{(0)} &= 0, \\ P'^{(0)} &= 0, \\ G'^{(0)} &= (\gamma\Pi + m)^{-1} = G_0, \\ \Gamma'^{(0)}_\mu(k) &= \gamma_\mu(k) = \gamma_\mu e^{-ikx}, \\ \mathfrak{G}'^{(0)}_{\mu\nu}(k, k') &= (\delta_{\mu\nu}/k^2) \delta(k - k'). \end{aligned} \quad (2.30)$$

To second order:

$$\begin{aligned} M'^{(2)} &= i(\alpha/4\pi^3) \int_r (dk)(dk') \gamma_\mu(-k) (\gamma\Pi + m)^{-1} \gamma_\nu(k') (\delta_{\mu\nu}/k^2) \delta(k - k') \\ &= i(\alpha/4\pi^3) \int_r (dk)(k^2)^{-1} \gamma_\mu [\gamma(\Pi - k) + m]^{-1} \gamma_\mu \end{aligned} \quad (2.31)$$

$$P'^{(2)} = -i(\alpha/4\pi^3) \text{tr} \int_r (dx)(dx') \gamma_\mu e^{-ikx} G'^{(0)}(x, x') \gamma_\nu e^{ik'x'} G'^{(0)}(x', x). \quad (2.32)$$

In deriving Eq. (31) we have used the fact that e^{ikx} acts like a displacement operator. That is

$$e^{ikx}G'[\Pi]e^{-ikx} = G'[\Pi - k]. \quad (2.33)$$

The fourth-order mass operator may be written

$$M'^{(4)} = M_{\mathbf{M}'} + M_{\mathbf{G}'} + M_{\mathbf{\Gamma}'} + M_{\mathbf{G}'}'. \quad (2.34)$$

Each term contains one of the quantities $\bar{M}_0$, G' , Γ' , and $\mathcal{G}'$ to second order and the others to lowest order. The four second-order ingredients are as follows:

$$[(1 + \bar{M}_0)^{-1}]^{(2)} = -\bar{M}_0^{(2)}, \quad (2.35)$$

$$G'^{(2)} = (\gamma\Pi + m + M'^{(2)})^{-1} - (\gamma\Pi + m) = -G_0 M'^{(2)} G_0, \quad (2.36)$$

$$\mathcal{G}'^{(2)} = -(1/k^2)P'^{(2)}(1/k^2). \quad (2.37)$$

The derivation of $\Gamma'^{(2)}$ is a bit more complicated. From Eqs. (9) and (31), we have

$$\begin{aligned} \Gamma_\mu^{(2)}(\xi) &= -i(\alpha/4\pi^3) \int (dk')(k'^2)^{-1} \gamma_\nu [\delta/\delta e A_\mu'(\xi)] G_0[\Pi - k'] \gamma_\nu \\ &= -i(\alpha/4\pi^3) \int (dk')(k'^2)^{-1} \gamma_\nu G_0[\Pi - k'] \gamma_\mu(\xi) G_0[\Pi - k'] \gamma_\nu, \end{aligned}$$

so that in momentum space

$$\Gamma_\mu^{(2)}(k) = -i(\alpha/4\pi^3) e^{-ikx} \int (dk')(k'^2)^{-1} \gamma_\nu G_0[\Pi - k - k'] \gamma_\mu G_0[\Pi - k'] \gamma_\nu. \quad (2.38)$$

We have made use of the fact that

$$[\delta/\delta e A_\mu'(\xi)] \Pi_\nu = -\delta_{\mu\nu}. \quad (2.39)$$

To renormalize $\Gamma_\mu(k)$ we must divide by $1 + \bar{M}_0$, which, to second order, means subtracting $\bar{M}_0^{(2)} \gamma_\mu(k)$.

In terms of these second-order ingredients, the four terms of the fourth-order mass operator are

$$M_{\mathbf{M}'} = -\bar{M}_0^{(2)} M'^{(2)}, \quad (2.40)$$

$$M_{\mathbf{G}'} = i(\alpha/4\pi^3) \int_r (dk)(k^2)^{-1} \gamma_\mu G'^{(2)}[\Pi - k] \gamma_\mu, \quad (2.41)$$

$$M_{\mathbf{\Gamma}'} = i(\alpha/4\pi^3) \int_r (dk)(k^2)^{-1} \gamma_\mu [\gamma(\Pi - k) + m]^{-1} e^{ikx} \Gamma_\mu^{(2)}(k) + M_{\mathbf{M}'}' \quad (2.42)$$

$$M_{\mathbf{G}'}' = i(\alpha/4\pi^3) \int_r (dk)(dk') \gamma_\mu [\gamma(\Pi - k) + m]^{-1} \gamma_\nu e^{ix(k-k')} \mathcal{G}_{\nu\mu}'^{(2)}(-k', -k). \quad (2.43)$$

To find the magnetic moment to second and fourth orders, we examine the expectation value of $M_F^{(2)} + M_F^{(4)}$ in a state whose wave function satisfies Eq. (1).

3. SECOND-ORDER CALCULATION

We introduce some simplifications in notation by writing A_μ and $F_{\mu\nu}$ for eA_μ' and $eF_{\mu\nu}'$. From the definition of Π in Eq. (2.8) we infer the operator commutation relation

$$[\Pi_\mu, \Pi_\nu] = iF_{\mu\nu}. \quad (3.1)$$

We have taken $F_{\mu\nu}$ to represent a constant field so that

$$[\Pi_\mu, F_{\nu\lambda}] = 0. \quad (3.2)$$

We define

$$\mathcal{F} = \frac{1}{2}\sigma_{\mu\nu}F_{\mu\nu} = (i/2)\gamma_\mu\gamma_\nu F_{\mu\nu}. \quad (3.3)$$

The two relativistically invariant scalar matrices, $\mathcal{F}$ and $\gamma\Pi$, obey the commutation relation

$$[\gamma\Pi, \mathcal{F}] = 2i\gamma F\Pi.$$

The second-order calculation contains, in short form, practically all of the features of the more lengthy fourth-order calculation. We consider the second-order mass operator as given in Eq. (2.31). Without the subscript r the integral would diverge for large k . Upon making the subtractions indicated by r , we would find that the integral would now converge for large k , but would diverge in the region of small k . To handle these difficulties in a consistent manner, we introduce two cutoffs: the ultraviolet cutoff λm^2 ($\lambda \rightarrow \infty$), and the infrared cutoff ϵm^2 ($\epsilon \rightarrow 0$). These are incorporated into the photon Green's function, $\mathcal{G}$, so that ϵm^2 behaves like a photon mass and $\lambda m^2/(k^2 + \lambda m^2)$ multiplies $\mathcal{G}$. That is, we use a new $\mathcal{G}(\lambda, \epsilon)$ given by

$$(k^2 + \epsilon m^2 + P)\mathcal{G}(\lambda, \epsilon) = \lambda m^2/(k^2 + \lambda m^2),$$

where $\lim_{\epsilon \rightarrow 0, \lambda \rightarrow \infty} \mathcal{G}(\lambda, \epsilon) = \mathcal{G}$. Then (2.31) takes the form

$$M'^{(2)} = \frac{i\alpha}{4\pi^3} \int_r (dk) \frac{1}{k^2 + \epsilon m^2} \frac{\lambda m^2}{k^2 + \lambda m^2} \gamma_\mu \frac{1}{\gamma(\Pi - k) + m} \gamma_\mu. \quad (3.4)$$

This type of integral will occur throughout the calculation. It is characterized by an integration over momenta k , and a sum over polarizations μ . The major complication is the presence of the γ -matrices and the variable k in the several denominators.

To eliminate the γ -matrices in favor of the $\mathfrak{F}$ -matrix, we rationalize the denominator:

$$[\gamma(\Pi - k) + m]^{-1} = [m - \gamma(\Pi - k)]\{m^2 - [\gamma(\Pi - k)]^2\}^{-1}.$$

By virtue of the anticommutativity of the γ 's and Eqs. (1) and (3) we have

$$[\gamma(\Pi - k)]^2 = \mathfrak{F} - (\Pi - k)^2 \quad (3.5)$$

so that

$$\begin{aligned} M'^{(2)} = \frac{i\alpha}{4\pi^3} \int_r (dk) \frac{1}{k^2 + \epsilon m^2} \frac{\lambda m^2}{k^2 + \lambda m^2} \gamma_\mu [m - \gamma(\Pi - k)] \\ \times \frac{1}{m^2 + (\Pi - k)^2 - \mathfrak{F}} \gamma_\mu. \end{aligned} \quad (3.6)$$

A simple method of doing the k -integration involves the introduction of integrals over auxiliary variables, by means of which the various denominators can be combined into a single factor. The causal Green's functions $G(x, x')$ and $\mathfrak{G}(\xi, \xi')$ are such that their Fourier expansions contain only positive frequencies for $x_0 > x'_0$ and $\xi_0 > \xi'_0$; and only negative frequencies for $x_0 < x'_0$ and $\xi_0 < \xi'_0$. These boundary conditions are made explicit in the momentum representation of the Green's functions $1/(\gamma\Pi + m)$ and $1/k^2$, by assigning an infinitesimal negative imaginary part to the denominators, which may be thought of as an addition to the electron mass m . It is seen from (6) that each of the denominators is thereby given the appropriate imaginary part.

We make use of the identity

$$1/a = i \int_0^\infty ds e^{-isa}, \quad (3.7)$$

which is valid provided that a possesses a negative imaginary part. We apply (7) to the three denominators of (6) and obtain

$$\begin{aligned} M'^{(2)} = (\alpha/4\pi^3) \lambda m^2 \int_0^\infty ds \int_0^\infty dt \int_0^\infty dq \\ \times \int_r (dk) \exp[-it(k^2 + \epsilon m^2) - iq(k^2 + \lambda m^2)] \gamma_\mu [m - \gamma(\Pi - k)] \\ \times \exp\{-is[m^2 + (\Pi - k)^2 - \mathfrak{F}]\} \gamma_\mu. \end{aligned} \quad (3.8)$$

The exponential containing the sum, $m^2 + (\Pi - k)^2 - \mathfrak{F}$, may be decomposed into a product of the individual exponentials, since the terms in this sum commute with one another. We are interested in terms linear in F and therefore may disregard any expressions of higher degree. We write, accordingly,

$$\exp is\mathfrak{F} = 1 + is\mathfrak{F} \quad (3.9)$$

to obtain

$$\begin{aligned}
 M'^{(2)} &= (\alpha/4\pi^3)\lambda m^2 \int_0^\infty ds \int_0^\infty dt \int_0^\infty dq \\
 &\times \int_r (dk) \exp[-im^2(s + t\epsilon + q\lambda)] \gamma_\mu [m + \gamma(\Pi - k)](1 + is\mathfrak{F})\gamma_\mu \quad (3.10) \\
 &\exp[-is(\Pi - k)^2 - i(t + q)k^2].
 \end{aligned}$$

The indicated sum over polarizations can now be carried out. The only machinery needed is the anticommutation relation of the γ -matrices. It follows that

$$\begin{aligned}
 \gamma_\mu \gamma_\mu &= -4, \\
 \gamma_\mu \gamma_\nu \gamma_\mu &= 2\gamma_\nu, \\
 \gamma_\mu \gamma_\nu \gamma_\lambda \gamma_\mu &= 4\delta_{\nu\lambda}, \\
 \gamma_\mu \gamma_\nu \gamma_\lambda \gamma_\sigma \gamma_\mu &= 2\gamma_\sigma \gamma_\lambda \gamma_\nu.
 \end{aligned} \quad (3.11)$$

The relations

$$\begin{aligned}
 \gamma_\mu \mathfrak{F} \gamma_\mu &= 0, \\
 \gamma_\mu \gamma_\nu \mathfrak{F} \gamma_\mu &= -2\mathfrak{F} \gamma_\nu
 \end{aligned} \quad (3.12)$$

are consequences of the fact that $F_{\mu\nu}$ is antisymmetric. The sum over polarizations for any expression of first degree in the field may be derived from these formulas. In the present example, we have

$$\gamma_\mu [m - \gamma(\Pi - k)](1 + is\mathfrak{F})\gamma_\mu = -4m - 2\gamma(\Pi - k) + 2is\mathfrak{F}\gamma(\Pi - k). \quad (3.13)$$

The k -integration has been reduced to an essentially Gaussian integral. Such an integral is normally computed by completing the square in the exponent with respect to k . Since we are dealing with noncommuting operators (for example $[\Pi_1 - k_1]^2$ and $\gamma_2[\Pi_2 - k_2]$), it is necessary to investigate the process of integration in some detail. It will turn out, that to first degree in the field, only slight modifications must be made in the normal procedure.

We consider the integral

$$I = \int (dk) [1 + c_1 k_\mu + c_2 k_\nu k_\lambda + c_3 k_\sigma k_\alpha k_\beta] \exp[-is(k^2 - 2ak\Pi + b\Pi^2)]. \quad (3.14)$$

To complete the square in the exponent we write

$$k^2 - 2ak\Pi + b\Pi^2 = (k - a\Pi)^2 + (b - a^2)\Pi^2.$$

We note that in any expression in which $F_{\mu\nu}$ appears or in which there is a commutator $[\Pi_\mu, \Pi_\nu]$, the remaining Π 's can be freely commuted with one another

(to first degree in F). Thus we may apply the formula

$$e^{(A+B)} = e^A e^B e^{-1/2[A, B]}$$

(in which it is assumed that $[A, [A, B]] = [B, [A, B]] = 0$), to our exponential and obtain

$$\begin{aligned} & \exp\{-is(k - a\Pi)^2 + \Pi^2(b - a^2)\} \\ &= \textstyle\frac{1}{2}\{\exp[-is(k - a\Pi)^2], \exp[-is(b - a^2)\Pi^2]\}. \end{aligned}$$

We have used the fact that the exponential of the commutator can be expanded to first degree in F . In the same vein we may write

$$\begin{aligned} & \exp\{-is[(k - a\Pi)^2 + \Pi^2(b - a^2)]\} \\ &= \textstyle\frac{1}{2}\{\exp[-is\Pi^2(b - a^2)] \exp[-is(k_1 - a\Pi_1)^2] \exp[-is(k_2 - a\Pi_2)^2] \\ & \quad \times \exp[-is(k_3 - a\Pi_3)^2] \exp[is(k_0 - a\Pi_0)^2]\}_{+\Pi}. \end{aligned}$$

where the notation $\{\}_{+\Pi}$ indicates that an average is to be taken of the quantity in braces and the same quantity with the order of Π -factors reversed. (Any γ -matrices that appear are *not* to be symmetrized in this manner.)

Each of the factors $k_\mu, k_\nu, \dots$ is now placed next to the exponential containing the same component of k . The integral can then be expressed as a product of four integrals of the form

$$I' = \int_{-\infty}^{\infty} dk_1(1 + c_1'k_1 + c_2'k_1^2 + c_3'k_1^3) \exp[-is(k_1 - a\Pi_1)^2], \quad (3.15)$$

each of which involves only a single component of Π , which, in turn, has no other operator with which not to commute. We can then apply ordinary methods of integration to I' . We let $k_1 - a\Pi_1 = \bar{k}_1$ so that the range of $\bar{k}_1$ also extends from $-\infty$ to ∞ . Terms of odd degree in $\bar{k}_1$ will integrate to zero since they transform to their negatives under the reflection $\bar{k}_1 \rightarrow -\bar{k}_1$. We are left with

$$\begin{aligned} I' &= \int_{-\infty}^{\infty} d\bar{k}_1[1 + c_1'a\Pi_1 + c_2'(a^2\Pi_1^2 + \bar{k}_1^2) + c_3'(a^3\Pi_1^3 + 3a\bar{k}_1^2\Pi_1)] \\ & \quad \times \exp(-is\bar{k}_1^2). \end{aligned} \quad (3.16)$$

Integrating by parts, we obtain

$$\int_{-\infty}^{\infty} d\bar{k}_1 \bar{k}_1^2 \exp[-is\bar{k}_1^2] = -(i/2s) \int_{-\infty}^{\infty} d\bar{k}_1 \exp(-is\bar{k}_1^2)$$

so that

$$\begin{aligned} I' &= [1 + c_1'a\Pi_1 + c_2'(a^2\Pi_1^2 - \textstyle\frac{1}{2}is^{-1}) + c_3'(a^3\Pi_1^3 - \textstyle\frac{3}{2}ia\Pi_1s^{-1})] \\ & \quad \times \int_{-\infty}^{\infty} d\bar{k}_1 \exp(-is\bar{k}_1^2). \end{aligned} \quad (3.17)$$

Still to be determined is the product of the four Gaussian integrals

$$\int_{-\infty}^{\infty} d\bar{k}_1 \exp(-is\bar{k}_1^2) \int_{-\infty}^{\infty} d\bar{k}_2 \exp(-is\bar{k}_2^2) \int_{-\infty}^{\infty} d\bar{k}_3 \exp(-is\bar{k}_3^2) \int_{-\infty}^{\infty} d\bar{k}_0 \exp(is\bar{k}_0^2).$$

The first three are all the same while the fourth is their complex conjugate. Thus the value of their product is

$$(\pi/is)^{1/2}(\pi/is)^{1/2}(\pi/is)^{1/2}[(\pi/is)^{1/2}]^* = (\pi/is)(\pi/s) = -i(\pi^2/s^2).$$

When we put together the four integrals of type I' , we obtain

$$\begin{aligned} I = & -i(\pi^2/s^2) \frac{1}{2} \{ [1 + c_1 a \Pi_\mu + c_2 (a^2 \Pi_\nu \Pi_\lambda - \frac{1}{2} is^{-1} \delta_{\mu\nu}) \\ & + c_3 (a^3 \Pi_\sigma \Pi_\alpha \Pi_\beta - \frac{1}{2} is^{-1} a [\Pi_\sigma \delta_{\alpha\beta} + \Pi_\alpha \delta_{\sigma\beta} + \Pi_\beta \delta_{\alpha\sigma}])] \\ & \times \exp[-is\Pi^2(b - a^2)] \}_{+\Pi}. \end{aligned} \quad (3.18)$$

We have shown that in the presence of the noncommuting operators Π_μ , Π_ν , the k -integrations, to first degree in F , are the same as for ordinary numbers, except that the result must be symmetrized in the manner indicated. A term which is of first degree in F before integration need not be symmetrized at all.

The square is completed in the exponent of (10) by writing

$$\begin{aligned} s(\Pi - k)^2 + (t + q)k^2 \\ = (s + t + q)[k - s\Pi/(s + t + q)]^2 + \Pi^2 s(t + q)/(s + t + q). \end{aligned}$$

The translation $\bar{k} = k - [s\Pi/(s + t + q)]$, and subsequent integration over $\bar{k}$ yield

$$\begin{aligned} M'^{(2)} = & -i(\alpha/4\pi)\lambda m^2 \int_0^\infty ds' \int_0^1 du (1 - u) \int_0^1 dw \\ & \times \exp\{-is'm^2[u + (1 - u)w\epsilon + (1 - u)(1 - w)\lambda]\} \\ & \times [(-4m - 2\gamma\Pi + 2is'u(1 - u)\mathfrak{F}\gamma\Pi) \exp[-is'\Pi^2 u(1 - u)] \\ & + \frac{1}{2}u\{2\gamma\Pi, \exp[-is'\Pi^2 u(1 - u)]\}] \end{aligned} \quad (3.19)$$

where the new variables s' , u , and w are given by

$$\begin{aligned} s' &= s + t + q, \\ u &= s/(s + t + q), \\ w &= t/(t + q). \end{aligned} \quad (3.20)$$

Since we intend to take matrix elements of $M'^{(2)}$ in a state for which $\gamma\Pi$ has a definite value, we use (5) to replace Π^2 by $(\gamma\Pi)^2$ in the exponentials of (19).

Also, to first degree in F we have $\exp [-is'u(1-u)\mathfrak{F}] = 1 - is'u(1-u)\mathfrak{F}$ so that

$$\begin{aligned} M'^{(2)} = & -i(\alpha/4\pi)\lambda m^2 \int_0^\infty ds \int_0^1 du(1-u) \int_0^1 dw \exp \{-im^2s \\ & \times [u + (1-u)w\epsilon + (1-u)(1-w)\lambda]\} \\ & \times [-4m - 2(1-u)\gamma\Pi + 4isu(1-u)m\mathfrak{F} + isu(1-u)(2-u)\{\gamma\Pi, \mathfrak{F}\}] \\ & \times \exp [isu(1-u)(\gamma\Pi)^2]. \end{aligned} \quad (3.21)$$

We have used the fact that $[\mathfrak{F}, (\gamma\Pi)^2] = 0$ and have dropped the prime from s' .

Equation (7) together with its generalization

$$i^{(n+1)} \int_0^\infty ds s^n e^{-ias} = n! a^{-(n+1)} \quad (3.22)$$

are used to obtain

$$\begin{aligned} M'^{(2)} = & \frac{\alpha}{4\pi} \lambda m^2 \int_0^1 du(1-u) \int_0^1 dw \\ & \times \left\{ \frac{4m + 2(1-u)\gamma\Pi}{m^2[u + (1-u)w\epsilon + (1-u)(1-w)\lambda] - u(1-u)(\gamma\Pi)^2} \right. \\ & \left. - \frac{4mu(1-u)\mathfrak{F} + u(1-u)(2-u)\{\mathfrak{F}, \gamma\Pi\}}{\{m^2[u + (1-u)w\epsilon + (1-u)(1-w)\lambda] - u(1-u)(\gamma\Pi)^2\}^2} \right\} \\ = & \frac{\alpha}{4\pi} \int_0^1 du \left\{ [4m + 2(1-u)\gamma\Pi] \log \frac{(1-u)\lambda m^2}{m^2[u + (1-u)\epsilon] - u(1-u)(\gamma\Pi)^2} \right. \\ & \left. - \frac{4mu(1-u)\mathfrak{F} + u(1-u)(2-u)\{\mathfrak{F}, \gamma\Pi\}}{m^2[u + (1-u)\epsilon] - u(1-u)(\gamma\Pi)^2} \right\} + O(1/\lambda). \end{aligned} \quad (3.23)$$

The second-order magnetic moment, $\mu^{(2)}$, is to be extracted from the field-dependent part of (23):

$$M_F^{(2)} = -\frac{\alpha}{4\pi} \int_0^1 du(1-u)u \frac{4m\mathfrak{F} + (2-u)\{\mathfrak{F}, \gamma\Pi\}}{m^2[u + (1-u)\epsilon] - u(1-u)(\gamma\Pi)^2} \quad (3.24)$$

$$\begin{aligned} (\Delta_F m)^{(2)} &= \langle \bar{\psi}_A | M_F^{(2)} | \psi_A \rangle \\ &= -\frac{\alpha\langle \mathfrak{F} \rangle}{4\pi m} \int_0^1 duu(1-u) \frac{2u}{u^2 + (1-u)\epsilon} \\ &= -\alpha\langle \mathfrak{F} \rangle/4\pi m + O(\sqrt{\epsilon}) \end{aligned} \quad (3.25)$$

where $\langle \mathfrak{F} \rangle = \langle \bar{\psi}_A | \mathfrak{F} | \psi_A \rangle$. Thus the second-order moment is

$$\mu^{(2)} = \alpha/2\pi. \quad (3.26)$$

We list the steps we have taken with the mass operator in obtaining this result, for they are to be repeated in the fourth-order calculation.

1. Rationalization of the denominators.
2. Exponentiation of the denominators.
3. Summation over polarizations μ .
4. Integration over momenta k by completing the square in a Gaussian integral and symmetrizing the result appropriately.
5. Introduction of new variables of integration such that only one integral extends from 0 to ∞ , while the others run from 0 to 1.
6. Integration over the variable running from 0 to ∞ , to convert the exponent back into a denominator.
7. Evaluation of the expectation value for a state with $\gamma\Pi = -m$.
8. Integration over the remaining auxiliary variables.

4. FOURTH-ORDER CALCULATION

A. CONTRIBUTION FROM THE SECOND-ORDER ELECTRON GREEN'S FUNCTION

To find the second-order electron Green's function as given in (2.36), we must perform the subtractions indicated by r on the integral in (3.23). Integration by parts yields

$$M_0^{(2)} = \frac{\alpha}{4\pi} \left\{ (4m + \gamma\Pi) \log \lambda - 4m - \frac{1}{2} \gamma\Pi + \int_0^1 du \frac{4mu + \gamma\Pi(2u - u^2)}{m^2[u + \epsilon(1 - u)] - u(1 - u)(\gamma\Pi)^2} [m^2(1 - \epsilon) - (1 - 2u)(\gamma\Pi)^2] \right\} \quad (4.1)$$

for the field-independent term. The expansion in powers of $(\gamma\Pi + m)$ is then

$$M_0^{(2)} = \frac{\alpha}{4\pi} \left\{ 3m \log \lambda - \frac{7}{2} m + (\gamma\Pi + m) \left(\log \lambda - \frac{1}{2} \right) + m \int_0^1 du \frac{(2u + u^2)(2u - \epsilon)}{u^2 + \epsilon(1 - u)} + (\gamma\Pi + m) \int_0^1 du \right. \\ \left. \times \frac{-4u^3 + 2u^4 - 2u^5 + \epsilon(4u - 2u^2 - 8u^3 + 5u^4) + \epsilon^2(-2u + 3u^2 - u^3)}{[u^2 + \epsilon(1 - u)]^2} \right\} + M_1^{(2)}. \quad (4.2)$$

We place $\epsilon = 0$ in those integrals which converge in this limit. The forms taken by the integrals which diverge or are indeterminate as $\epsilon \rightarrow 0$ are found by substituting $u = u'\sqrt{\epsilon}$, and dropping terms of higher order in $\sqrt{\epsilon}$. The result is

$$M_1^{(2)} = M_0^{(2)} - Bm - \bar{M}_0^{(2)}(\gamma\Pi + m), \quad (4.3)$$

where

$$B = (3\alpha/4\pi)(\log \lambda + \frac{1}{2})$$

and

$$\tilde{M}_0^{(2)} = (\alpha/4\pi)(\log \lambda + \frac{9}{2} + 2 \log \epsilon). \quad (4.4)$$

The Green's function is then

$$\begin{aligned} G'^{(2)} &= \{\gamma\Pi + m + M^{(2)} - Bm - (\gamma\Pi + m)\tilde{M}_0^{(2)}\}^{-1} - (\gamma\Pi + m)^{-1} \\ &= \left\{ \frac{1}{\gamma\Pi + m + M^{(2)}} - \frac{1}{\gamma\Pi + m} \right\} + \frac{\tilde{M}_0^{(2)}}{\gamma\Pi + m} \\ &\quad + \left\{ \frac{1}{\gamma\Pi + m(1-B)} - \frac{1}{\gamma\Pi + m} \right\} \\ &= G'^{(2)a} + G'^{(2)b} + G'^{(2)c}. \end{aligned} \quad (4.5)$$

Except for a multiplicative factor, $G'^{(2)b}$ is the same as G_0 and thus gives rise to a fourth-order moment

$$\begin{aligned} \mu_{\sigma}^{(4)b} &= \tilde{M}_0^{(2)} \mu^{(2)} \\ &= (\alpha^2/\pi^2)(\frac{1}{8} \log \lambda + \frac{9}{16} + \frac{1}{4} \log \epsilon). \end{aligned} \quad (4.6)$$

The first term in $G'^{(2)c}$ is the same as G_0 except that m has been replaced by $m(1-B)$. Equation (3.24) then tells us

$$\begin{aligned} (\Delta_F m)^{(4)c}_{\sigma} &= \left\langle \bar{\psi}_A \left| -\frac{\alpha}{4\pi} \int_0^1 du u(1-u) \right. \right. \\ &\quad \times \frac{4m\mathfrak{F}(1-B) + (2-u)\{\mathfrak{F}, \gamma\Pi\}}{m^2(1-B)^2[u + (1-u)\epsilon] - u(1-u)(\gamma\Pi)^2} - M_F^{(2)} \left. \right| \psi_A \rangle \\ &= -\frac{\alpha B \langle \mathfrak{F} \rangle}{4\pi m} \int_0^1 du u(1-u) \left\{ -\frac{4}{u^2 + \epsilon(1-u)} + \frac{4u[u + \epsilon(1-u)]}{[u^2 + \epsilon(1-u)]^2} \right\} \\ &= \alpha B \langle \mathfrak{F} \rangle / 2\pi m \\ \mu_{\sigma}^{(4)c} &= (\alpha^2/\pi^2)(-\frac{3}{4} \log \lambda - \frac{3}{8}). \end{aligned} \quad (4.7)$$

We rationalize and exponentiate the denominator of $G'^{(2)a}[\Pi]$ by writing it as follows:

$$\begin{aligned} G'^{(2)a}[\Pi] &= (m - \gamma\Pi + M^{(2)})\{m^2 - (\gamma\Pi)^2 + [\gamma\Pi, M^{(2)}] + 2mM^{(2)}\}^{-1} \\ &\quad - (m - \gamma\Pi)\{m^2 - (\gamma\Pi)^2\}^{-1} \\ &= i \int_0^\infty ds \{M^{(2)} - is(m - \gamma\Pi)(2mM^{(2)} + [\gamma\Pi, M^{(2)}])\} \\ &\quad \times \exp \{-is[m^2 - (\gamma\Pi)^2]\}. \end{aligned} \quad (4.8)$$

We place this expression into Eq. (2.41), using Eq. (3.19) for $M^{(2)}$, and obtain

$$\begin{aligned}
 M_a^a = & -i(\alpha^2/16\pi^4)\lambda^2 m^4 \int_0^\infty ds \int_0^\infty ds' \int_0^\infty dt \int_0^\infty dq \int_0^1 du \int_0^1 dw \\
 & \times \int (dk)(1-u) \exp \{-im^2[s + s'u + s'(1-u)w\epsilon \\
 & + s'(1-u)(1-w)\lambda] - it(k^2 + \epsilon m^2) - iq(k^2 + \lambda m^2)\} \\
 & \times \gamma_\mu N_1 \gamma_\mu \exp \{-i(\Pi - k)^2[s + s'u(1-u)]\}.
 \end{aligned} \tag{4.9}$$

The expression N_1 is a polynomial function of the various operators, matrices, variables and constants, the details of which will not be reproduced here. The basic structure of the integral is contained, not in N_1 , but in the accompanying exponentials.

Performing the indicated sum over polarizations μ , we obtain a new polynomial N_2 . The exponent whose square is to be completed involves

$$\begin{aligned}
 & k^2[s + s'u(1-u) + t + q] - 2[s + s'u(1-u)]k\Pi + [s + s'u(1-u)]\Pi^2 \\
 = & [s + s'u(1-u) + t + q]\{k - \Pi[s + s'u(1-u)]/[s + s'u(1-u) + t + q]\}^2 \\
 & + \Pi^2[s + s'u(1-u)](t + q)/[s + s'u(1-u) + t + q].
 \end{aligned}$$

The integration proceeds as indicated in (3.18). The translation is

$$\bar{k} = k - \Pi[s + s'u(1-u)]/[s + s'u(1-u) + t + q].$$

We obtain

$$\begin{aligned}
 M_a^a = & -(\alpha^2/16\pi^2)\lambda^2 m^4 \int_0^\infty ds \int_0^\infty ds' \int_0^\infty dt \int_0^\infty dq \int_0^1 du \int_0^1 dw (1-u) \\
 & \times [s + s'u(1-u) + t + q]^{-2} \\
 & \times \exp \{-im^2[s + s'u + \epsilon t + \epsilon s'(1-u)w + \lambda q + \lambda s'(1-u)(1-w)]\} \\
 & \times (N_3, \exp \{-i\Pi^2[s + s'u(1-u)](t + q)/[s + s'u(1-u) + t + q]\})
 \end{aligned} \tag{4.10}$$

where the parentheses indicate the appropriate symmetrization of the exponential with terms of N_3 .

We change variables:

$$\begin{aligned}
 p &= s + s'u(1-u) + t + q, \\
 x &= [s + s'u(1-u)]/[s + s'u(1-u) + t + q], \\
 v &= s'u(1-u)/[s + s'u(1-u)], \\
 w' &= t/(t + q).
 \end{aligned} \tag{4.11}$$

In order to express everything in terms of $\gamma\Pi$ and $\mathfrak{F}$, we use

$$\exp[-ipx(1-x)\Pi^2] = [1 - ipx(1-x)\mathfrak{F}] \exp[ipx(1-x)(\gamma\Pi)^2]$$

and unsymmetrize the quantity in parentheses, thereby introducing extra terms into N_3 :

$$\begin{aligned} M_G^a = & -(\alpha^2/16\pi^2)\lambda^2 m^4 \int_0^\infty dp \int_0^1 du \int_0^1 dv \int_0^1 dx \int_0^1 dw \\ & \times \int_0^1 dw' p u^{-1} x(1-x) \exp \left\{ -im^2 p \left[x \left(1 - v + \frac{v}{1-u} \right) \right. \right. \\ & \left. \left. + \epsilon x \frac{vw}{u} + \lambda x \frac{v(1-w)}{u} + \epsilon(1-x)w' + \lambda(1-x)(1-w') \right] \right\} \\ & \times N_4 \exp[ipx(1-x)(\gamma\Pi)^2]. \end{aligned} \quad (4.12)$$

We are interested in the field-dependent part evaluated for $\gamma\Pi = -m$:

$$\begin{aligned} (\Delta_F m)^{(4)}_G^a = & \langle \bar{\psi}_A | M_{FG}^a | \psi_A \rangle \\ = & -(\alpha^2/16\pi^2) \langle \mathfrak{F} \rangle \lambda^2 m^4 \int_0^\infty dp \int_0^1 du \int_0^1 dv \int_0^1 dx \int_0^1 dw \\ & \int_0^1 dw' p^2 u^{-1} x^2 (1-x) \\ & \times \exp \left\{ -ipx m^2 \left[x - v + \frac{v}{1-u} + \epsilon \frac{vw}{u} + \lambda \frac{v(1-w)}{u} \right] \right. \\ & \left. - ip(1-x) m^2 [\epsilon w' + \lambda(1-w')] \right\} \\ & \times \{ m[-16i(1-x) - 4i(1-u)(1-x)^2 + 32i(1-u)(1-v)x(1-x) \\ & - 16i(1-u)(1-v) - 4i(1-u)(1-x) + 4iv(2-u)(1-x) \\ & + 16i(1-v)x(2-x)(1-u)] + m^3 px(1-x)[-32(1-v) \\ & - 8(1-v)(1-x)(1-u) + 16(1-v)(1-x) \\ & + 16(1-v)(1-x)^2(1-u) - 8(1-v)(1-u) + 16(1-v) \\ & + 8v(1-v)(2-u) - 16v(1-v)] \}. \end{aligned} \quad (4.13)$$

The integrations with respect to w , w' , and p yield

$$(\Delta_F m)^{(4)}_G^a = (\alpha^2/4\pi^2 m) \langle \mathfrak{F} \rangle (A_{\epsilon\epsilon} - A_{\lambda\epsilon} - A_{\epsilon\lambda} + A_{\lambda\lambda}), \quad (4.14)$$

where

$$\begin{aligned}
 A_{\phi\psi} = & \int_0^1 du \int_0^1 dv \int_0^1 dx v^{-1} \\
 & \times \left\{ \frac{2u^2(1-u)^2x^2(1-x)(1-v)[uv + x(5-3u-2x+2ux)]}{[u(1-u)x^2 + u^2xv + (1-u)xv\phi + u(1-u)(1-x)\psi]^2} \right. \\
 & \quad \left. + u(1-u)x(1-x) \right. \\
 & \quad \left. \times \frac{-10 + 6u + 13x - 13ux + v(6-5u-12x+12ux)}{u(1-u)x^2 + u^2xv + (1-u)xv\phi + u(1-u)(1-x)\psi} \right\}. \quad (4.15)
 \end{aligned}$$

We are interested in the leading terms of this expression in the limits $\epsilon \rightarrow 0$ and $\lambda \rightarrow \infty$. The presence of λ in the denominators of $A_{e\lambda}$, $A_{\lambda e}$, and $A_{\lambda\lambda}$ would seem to indicate that these terms are of order $1/\lambda$. But in the region of integration where the coefficient of λ is small, the λ term in the denominator may no longer predominate. In $A_{\lambda e}$, if $v \sim 1/\lambda$, $dv \sim 1/\lambda$, then $A_{\lambda e} \sim dv/v \sim 1$ and not $\sim 1/\lambda$. On the other hand, in $A_{e\lambda}$ if $(1-x)$ or $(1-u) \sim 1/\lambda$, there are sufficient powers of $(1-u)$ and $(1-x)$ in the numerator to cause $A_{e\lambda}$ to vanish at least as fast as $1/\lambda$. The same applies to $A_{\lambda\lambda}$. Thus we have

$$(\Delta_F m)^{(4)}_G^a = (\alpha^2/4\pi^2 m) \langle \mathcal{F} \rangle (A_{e\epsilon} - A_{\lambda\epsilon}). \quad (4.16)$$

By means of the substitution $v = v'/\lambda$ in $A_{\lambda e}$ and the dropping of terms of higher order in $1/\lambda$, we find

$$\begin{aligned}
 A_{e\epsilon} - A_{\lambda\epsilon} = & \int_0^1 du \int_0^1 dx x(1-x) \\
 & \times \left\{ \frac{9(1-u)x^3 + \epsilon(1-x)(-10 + 6u + 13x - 13ux)}{[x^2 + \epsilon(1-x)]^2} \right. \\
 & \quad \times \log \frac{\lambda x(1-u)}{[ux + \epsilon(1-u)][u+x-ux]} \\
 & \quad \left. + \frac{2u(1-u)x^2(5-3u-2x+2ux)}{[x^2 + \epsilon(1-x)][u+x-ux][ux + \epsilon(1-u)]} \right\} \quad (4.17) \\
 & + \int_0^1 du \int_0^1 dx \int_0^1 dv \left\{ \frac{2x^2u^3(1-u)^2}{(u+x-ux)^2[ux + \epsilon(1-u)]^2} \right. \\
 & \quad \left. + \frac{(1-u)(1-x)(6-5u-12x+12ux)}{x(1-u) + vu} \right. \\
 & \quad \left. - (1-u)^2 \frac{2uv(1-x) + 2ux + 2x(1-x)(5-3u-2x+2ux)}{x(1-u) + uv} \right\}.
 \end{aligned}$$

We have set $\epsilon = 0$ in terms which converge in that limit. For those terms which are divergent or indeterminate when $\epsilon = 0$ we use the substitution $x = \sqrt{\epsilon} x'$ and reject terms of higher order in $\sqrt{\epsilon}$. The integrations are straightforward and yield

$$A_{\epsilon\epsilon} - A_{\lambda\epsilon} = -\frac{5}{4} \log \lambda - 1 - \frac{1}{2} \log \epsilon. \quad (4.18)$$

Using Eqs. (16), (6) and (7), we obtain

$$\mu^{(4)}_G{}^a = (\alpha^2/\pi^2)(\frac{5}{8} \log \lambda + \frac{1}{2} + \frac{1}{4} \log \epsilon) \quad (4.19)$$

and

$$\mu^{(4)}_G = \mu^{(4)}_G{}^a + \mu^{(4)}_G{}^b + \mu^{(4)}_G{}^c = (\alpha^2/\pi^2)(\frac{11}{16} + \frac{1}{2} \log \epsilon). \quad (4.20)$$

B. CONTRIBUTION FROM THE SECOND-ORDER PHOTON GREEN'S FUNCTION

The second-order photon Green's function is given by Eq. (2.37). The fact that $\mathfrak{G}$ is vacuum expectation value quadratic in A_μ implies that $\mathfrak{G}$, and hence P' are even functions of the external field. In particular, P' possesses no term linear in F so that for our purposes it suffices to look at the zero-field part of P' . That is, we use the zero-field electron Green's function $(\gamma p + m)^{-1}$ in Eq. (2.32). This may be written as an integral over intermediate electron momenta

$$\begin{aligned} P'_{\mu\nu}{}^{(2)}(k, k') &= -i(\alpha/4\pi^3) \text{tr} \int_{\tau} (dp)(dp') \gamma_\mu \langle p' | e^{-ikx} | p \rangle \\ &\quad \times (\gamma p + m)^{-1} \gamma_\nu \langle p | e^{ik'x'} | p' \rangle (\gamma p' + m)^{-1}. \end{aligned} \quad (4.21)$$

Noting that

$$\begin{aligned} \langle p' | e^{-ikx} | p \rangle &= (2\pi)^{-4} \int (dx)(dx') e^{-ip'x} e^{-ikx} \delta(x - x') e^{ipx'} = \delta(p' - p + k) \\ \langle p | e^{ik'x} | p' \rangle &= \delta(p - p' - k') \end{aligned}$$

we may write

$$P'_{\mu\nu}{}^{(2)} = -\delta(k - k')(i\alpha/4\pi^3) \text{tr} \int_{\tau} (dp) \gamma_\mu (\gamma p + m)^{-1} \gamma_\nu [\gamma(p - k) + m]^{-1}. \quad (4.22)$$

Since $P_{\mu\nu}'$ describes the polarization current we must require that it satisfy the conservation condition

$$k_\mu P_{\mu\nu}' = 0. \quad (4.23)$$

The p -integration in Eq. (22) is similar to the k -integrations that we have already discussed, and in fact simpler. For there are no noncommuting operators. To maintain the validity of (23) a proper cutoff procedure must be used. Two such

procedures are described by Feynman (7) and Schwinger (4). The result is

$$P'_{\mu\nu}{}^{(2)} = \delta(k - k') \left(\delta_{\mu\nu} - \frac{k_\mu k_\nu}{k^2} \right) \frac{\alpha}{2\pi} k^2 \int_{s_0}^{\infty} \frac{ds}{s} \int_0^1 du (1 - u^2) \quad (4.24)$$

$$\times \exp [-ism^2 - \frac{1}{4}isk^2(1 - u^2)].$$

The projection operator $[\delta_{\mu\nu} - (k_\mu k_\nu/k^2)]$ guarantees the transverse character of $P_{\mu\nu}'$. The lower limit $s_0(\rightarrow 0)$ is the cutoff. Subtracting the first two powers of k^2 we find

$$P'_{\mu\nu}{}^{(2)} = - \left(\delta_{\mu\nu} - \frac{k_\mu k_\nu}{k^2} \right) \delta(k - k') (k^2)^2 \frac{\alpha}{12\pi m^2} \quad (4.25)$$

$$\times \int_0^1 du (3u^2 - u^4) \left[1 + \frac{k^2}{4m^2} (1 - u^2) \right]^{-1}$$

$$G'_{\mu\nu}{}^{(2)}(-k', -k) = \left(\delta_{\mu\nu} - \frac{k_\mu k_\nu}{k^2} \right) \delta(k - k') \frac{\alpha}{12\pi m^2} \quad (4.26)$$

$$\times \int_0^1 du \frac{3u^2 - u^4}{1 + (k^2/4m^2)(1 - u^2)}.$$

The longitudinal, $k_\mu k_\nu/k^2$, part of $G'_{\mu\nu}{}^{(2)}$ does not affect the evaluation of any observable quantities. Thus the part of the fourth-order mass operator contributed to by $G'^{(2)}$ is

$$M_g = i(\alpha^2/48\pi^4 m^2) \int_0^1 du (3u^2 - u^4) \int_r (dk) \gamma_\mu [\gamma(\Pi - k) + m]^{-1} \quad (4.27)$$

$$\times \gamma_\mu [1 + (k^2/4m^2)(1 - u^2)]^{-1}.$$

We do not need any cutoffs here because we are interested only in the F -dependent part of M , for which we obtain

$$M_{Fg} = -i \frac{\alpha^2}{12\pi^2} \int_0^1 du \frac{3u^2 - u^4}{1 - u^2} \int_0^\infty ds \quad (4.28)$$

$$\int_0^1 dx (1 - x)x \exp [-im^2sx - 4im^2s(1 - x)/(1 - u^2)]$$

$$\times [4m\mathcal{F} + (2 - x)\{\gamma\Pi, \mathcal{F}\}] \exp [isx(1 - x)(\gamma\Pi)^2].$$

The field-dependent self-energy is then

$$(\Delta_F m)^{(4)}_g = \langle \bar{\psi}_A | M_{Fg} | \psi_A \rangle$$

$$= -\frac{\alpha^2 \langle \mathcal{F} \rangle}{6\pi^2 m} \int_0^1 dx (1 - x) \int_0^1 du (3u^2 - u^4) \left[\left(\frac{2 - x}{x} \right)^2 - u^2 \right]^{-1} \quad (4.29)$$

$$= -\frac{\alpha^2 \langle \mathcal{F} \rangle}{6\pi^2 m} \left[\frac{119}{12} - \pi^2 \right]$$

so that

$$\mu^{(4)}_g = (\alpha^2/\pi^2)(119/36 - \pi^2/3). \quad (4.30)$$

C. CONTRIBUTION FROM THE SECOND-ORDER VERTEX OPERATOR

The vertex part of the mass operator contains two terms, as given in Eq. (2.42). The contribution of $M_{M'}$ to the magnetic moment is just

$$\mu^{(4)}_M = -\bar{M}_0^{(2)} \mu^{(2)}. \quad (4.31)$$

Using Eq. (4), we have

$$\mu^{(4)}_M = (\alpha^2/\pi^2)[-1/8 \log \lambda - 9/16 - 1/4 \log \epsilon]. \quad (4.32)$$

The unrenormalized second-order vertex operator is given by Eq. (2.38) together with the cutoff modifications:

$$\begin{aligned} e^{ikx} \Gamma_\mu^{(2)}(k) = & -\frac{i\alpha}{4\pi^3} \int (dk') \gamma_\nu \frac{1}{\gamma(\Pi - k - k') + m} \gamma_\mu \frac{1}{\gamma(\Pi - k') + m} \gamma_\nu \\ & \times \frac{1}{k'^2 + \epsilon m^2} \frac{\lambda m^2}{k'^2 + \lambda m^2}. \end{aligned} \quad (4.33)$$

The details of the integration are quite similar to those we have already discussed. We shall content ourselves here with describing the choice of a set of variables which will make the final parameter integrations as simple as possible. To this end we must examine the development of the exponentials in which we complete squares.

Upon introducing an auxiliary variable for each of the denominators in (33), we obtain an exponent (apart from a factor $-i$):

$$\begin{aligned} & m^2(r + s + t\epsilon + q\lambda) + r(\Pi - k - k')^2 + s(\Pi - k')^2 + (t + q)k'^2 \\ & = m^2(r + s + t\epsilon + q\lambda) \\ & + (r + s + t + q) \left[k' - \frac{(r + s)\Pi}{r + s + t + q} + \frac{rk}{r + s + t + q} \right]^2 \\ & + \frac{(r + s)(t + q)}{r + s + t + q} \Pi^2 - 2 \frac{r(t + q)}{r + s + t + q} \Pi k + \frac{r(s + t + q)}{r + s + t + q} k^2. \end{aligned} \quad (4.34)$$

With the introduction of the new variables

$$\begin{aligned} p &= r + s + t + q, \\ u &= (r + s)/(r + s + t + q), \\ v &= r/(r + s), \\ w &= t/(t + q), \end{aligned} \quad (4.35)$$

the exponent may be written

$$m^2 p \{ u + (1 - u)[\epsilon w + \lambda(1 - w)] \} + p(k' - u\Pi + uvk)^2 \\ + pu(1 - u)\Pi^2 - 2pu(1 - u)vk\Pi + puw(1 - w)k^2. \quad (4.36)$$

The translation $k' = \bar{k}' + u\Pi - kuv$ and subsequent integration over $\bar{k}'$ leaves the exponent without its second term.

The expression for $e^{ikx}\Gamma_\mu^{(2)}(k)$ thus obtained is inserted into Eq. (2.42). Three additional parameters, s' , t' , and q' , are needed for the new denominators. The additional contribution to the exponent of (36) will be

$$m^2[s' + t'\epsilon + q'\lambda] + s'(\Pi - k)^2 + (t' + q')k^2. \quad (4.37)$$

After we complete the square with respect to k , the entire exponent is brought into the form

$$m^2 \{ pu + s' + \epsilon[p(1 - u)w + t'] + \lambda[p(1 - u)(1 - w) + q'] \} \\ + [puw(1 - w) + s' + t' + q'] \left\{ k - \frac{[s' + pu(1 - u)v]\Pi}{puw(1 - w) + s' + t' + q'} \right\}^2 \\ + \Pi^2 \left\{ s' + pu(1 - u) - \frac{[s' + pu(1 - u)v]^2}{puw(1 - w) + s' + t' + q'} \right\}. \quad (4.38)$$

If we introduce the variables,

$$r' = puw(1 - w) + s' + t' + q', \\ x = [puw(1 - w) + s']/[puw(1 - w) + s' + t' + q'], \\ y = puw(1 - w)/[puw(1 - w) + s'], \\ w' = t'/(t' + q'), \quad (4.39)$$

then after integration over k expression (38) becomes

$$m^2 r' \left\{ \frac{xy}{v(1 - w)} + x(1 - y) + \epsilon \left[\frac{y(1 - u)wx}{w(1 - w)} + w'(1 - x) \right] \right. \\ \left. + \lambda \left[\frac{y(1 - u)x(1 - w)}{w(1 - w)} + (1 - w')(1 - x) \right] \right\} \\ + \Pi^2 r' x \left\{ (1 - y) + \frac{y(1 - u)}{v(1 - w)} - x \left[(1 - y) + \frac{y(1 - u)}{1 - w} \right] \right\}. \quad (4.40)$$

The next steps are to set $\Pi^2 = \mathfrak{F} - (\gamma\Pi)^2$, take the expectation value in a state for which $\gamma\Pi = -m$ and integrate over r' . If we confine ourselves to the F -dependent part of $M_\Gamma^{(4)}$, the integrations over w and w' can be performed and we obtain four terms, which arise from w and $w' = 0$ or 1, just as in Eq. (14). Methods

similar to these we used before must be called upon to extract the leading terms in ϵ and λ . We are then left with the following structure, now as a denominator:

$$\begin{aligned} \left[\frac{y}{v(1-uv)} + 1 - y \right] - \left\{ 1 - y + \frac{y(1-u)}{v(1-uv)} \right. \\ \left. - x \left[1 - y + \frac{y(1-u)}{1-uv} \right]^2 \right\} \\ = \frac{yu}{v(1-uv)} + x \left[1 - y + \frac{y(1-u)}{1-uv} \right]^2. \end{aligned} \quad (4.41)$$

At this point our integral is a sum of terms of rational form. The numerators are polynomials in u, v, x, y and the denominator is the one we have exhibited either as it stands, or squared, together with various factors like $(1-uv)$ or v . The whole works, however, is explicitly finite. All integrations go from 0 to 1.

The best procedure is to make additional changes of variables, such that expression (41) becomes factorable. Consider

$$I(v, x) = \int_0^1 dy \int_0^1 du f(u, y, v, x). \quad (4.42)$$

We make the substitution

$$y(1-v)u = (1-y')(1-uv) \quad (4.43)$$

so that

$$\begin{aligned} I(v, x) &= \int_0^1 du \int_{(1-u)/(1-uv)}^1 dy' \frac{1-uv}{u(1-v)} f[u, y(y', u, v), v, x] \\ &= \int_0^1 dy' \int_{(1-y')/(1-vy')}^1 du \frac{1-uv}{u(1-v)} f[u, y(y', u, v), v, x]. \end{aligned} \quad (4.44)$$

Our denominator is then

$$\frac{1-y'}{v(1-v)} + xy'^2 = \frac{1}{v(1-v)} [xy'^2 v(1-v) - y' + 1] \quad (4.45)$$

and the u -integration can be done forthwith.

Finally, the substitutions

$$\begin{aligned} \xi &= 1 - vxy'; \\ \eta &= (1 - vy')/(1 - vxy'); \quad \frac{\partial(y', x, v)}{\partial(\xi, \eta, \zeta)} = \frac{\xi^2 \eta}{(1 - \xi\eta)(1 - \xi\eta\zeta)} \\ \zeta &= (1 - y')/(1 - vy'); \end{aligned} \quad (4.46)$$

bring the denominator into the form

$$(1 - \xi\eta\zeta)^2(1 - \xi + \xi\zeta)(1 - \zeta)^{-1}(1 - \xi\eta)^{-1} \quad (4.47)$$

a product of factors each of which is linear in ξ , η , and ζ . The η integral is evaluated first and then, by means of several integrations by parts, the ζ and ξ integrations are performed. An advantage of this method is that each term is finite at every step. Some of the ξ integrals are Spence functions which are evaluated using the results of Sandham (8). For example, we have

$$\begin{aligned} -2 \int_0^1 d\xi [(2 - \xi)^{-1} \log \xi + \xi^{-1} \log (2 - \xi)] \log (1 - \xi) \\ = -2 \int_0^2 d\xi \xi^{-1} \log |1 - \xi| \log (2 - \xi) = -2^{1/4} \zeta(3) + \pi^2 \log 2, \end{aligned}$$

where $\zeta(3)$ is the Riemann-zeta function of 3.

The result is

$$\begin{aligned} \mu_{\Gamma}^{(4)} &= (\alpha^2/\pi^2) [1/4 \log \lambda - 3/2 + 5/12 \pi^2 + 3/4 \zeta(3) - 1/2 \pi^2 \log 2] + \mu_M^{(4)} \\ &= (\alpha^2/\pi^2) [1/8 \log \lambda - 33/16 + 5/12 \pi^2 + 3/4 \zeta(3) - 1/2 \pi^2 \log 2 - 1/4 \log \epsilon]. \end{aligned} \quad (4.48)$$

The total fourth-order moment is found by adding the contributions from (20), (30), (32), and (48):

$$\mu^{(4)} = \frac{\alpha^2}{\pi^2} \left[\frac{197}{144} + \frac{\pi^2}{12} + \frac{3}{4} \zeta(3) - \frac{1}{2} \pi^2 \log 2 \right] \quad (4.49)$$

$$= -0.328(\alpha^2/\pi^2). \quad (4.50)$$

APPENDIX—ANOTHER METHOD

The reader, after observing how the infrared and ultraviolet divergences have cancelled, may wonder if it is not possible to compute $M_{\Gamma} + M_{\sigma}$ directly. Indeed, it is possible and furthermore one can do the entire calculation without introducing infrared or ultraviolet cutoffs. Unfortunately, there apparently is more labor involved in this procedure than in extracting the effects of the cutoffs.

We have

$$M_{\sigma} + M_{\Gamma} = i(\alpha/4\pi^3) \int (dk) (k^2)^{-1} \gamma_{\mu} e^{ikx} [G_0 \Gamma_{\mu}'^{(2)}(k) + G'^{(2)} \gamma_{\mu} e^{-ikx}]. \quad (A.1)$$

Considering

$$\begin{aligned} \Gamma_{\mu}'(k) &= \int (dx) e^{-ikx} \Gamma_{\mu}'(x) = - \int (dx) e^{-ikx} [\delta/\delta A_{\mu}(x)] G'^{-1} \\ &= - \int (dx) (dk') e^{-ikx} [\delta A_{\nu}(k')/\delta A_{\mu}(x)] \{ [\delta/\delta A_{\nu}(k')] G'^{-1} \} \\ &= - \int (dx) (dk') e^{-ikx} e^{-ik'x} \{ [\delta/\delta A_{\mu}(k')] G'^{-1} \} \\ &= -(2\pi)^4 [\delta/\delta A_{\mu}(-k)] G'^{-1} \end{aligned} \quad (A.2)$$

we have, in terms of the notation

$$(2\pi)^4[\delta/\delta A_\mu(-k)] = \delta_A, \quad (\text{A.3})$$

$$\begin{aligned} \Gamma'_\mu{}^{(2)}(k) &= -\delta_A(G'^{-1})^{(2)} \\ &= \delta_A[G_0^{-1}G'^{(2)}G_0^{-1}] \\ &= \{\delta_A[G_0^{-1}G'^{(2)}]\}G_0^{-1} + G_0^{-1}G'^{(2)}\delta_A G_0^{-1} \\ &= -\{\delta_A[M'^{(2)}G_0]\}G_0^{-1} - G_0^{-1}G'^{(2)}\gamma_\mu e^{-ikx}. \end{aligned} \quad (\text{A.4})$$

When this expression is placed in (A.1) the second term will cancel with M_c , leaving

$$M_G + M_\Gamma = -i(\alpha/4\pi^3) \int (dk) (k^2)^{-1} \gamma_\mu e^{ikx} G_0 \{\delta_A[M'^{(2)}G_0]\} G_0^{-1}. \quad (\text{A.5})$$

We have already computed an expression for $M'^{(2)}$ in Eq. (3.23). But we need more. For consider the derivative

$$\begin{aligned} \delta_A F_{\lambda\sigma} &= -i\delta_A[\Pi_\lambda, \Pi_\sigma] \\ &= i[e^{-ikx}\delta_{\mu\lambda}, \Pi_\sigma] + i[\Pi_\lambda, e^{-ikx}\delta_{\mu\sigma}] \\ &= ie^{-ikx}[\delta_{\mu\lambda}k_\sigma - \delta_{\mu\sigma}k_\lambda]. \end{aligned} \quad (\text{A.6})$$

The effect is one of reducing the degree of F . Hence we would have to compute $M'^{(2)}$ to second degree in F to obtain a derivative correct to first degree. Furthermore, in such a calculation we could no longer make use of the fact that F is a constant field. For we observe that even though $[\Pi_\nu, F_{\lambda\sigma}] = 0$, the same is not true of the derivative:

$$\delta_A[\Pi_\nu, F_{\lambda\sigma}] = [\Pi_\nu, \delta_A F_{\lambda\sigma}] = -ie^{-ikx}k_\nu[\delta_{\mu\lambda}k_\sigma - \delta_{\mu\sigma}k_\lambda] \neq 0.$$

We note that as long as δ_A does not act directly on the field F , we may correctly use Eq. (3.23) for $M'^{(2)}$. Thus we may write

$$M'^{(2)} = M_1^{(2)} + M_F^{(2)} + \tilde{M}^{(2)} \quad (\text{A.7})$$

where the effect of $\tilde{M}^{(2)}$ appears only when δ_A acts directly on F .

Let us consider the contribution of $M_1^{(2)} + M_F^{(2)}$ in Eq. (5). We compute $M_1^{(2)}$ by making the subtractions indicated in Eq. (4.3). Letting $\epsilon = 0$, we find

$$\begin{aligned} M_1^{(2)} &= (\gamma\Pi + m)^2 \frac{\alpha}{4\pi} \int_0^1 \frac{du}{u} \frac{\gamma\Pi(-4 + 4u + u^2) + m(4 - 2u)}{m^2 - (1 - u)(\gamma\Pi)^2} \\ M_1^{(2)}G_0 &= M_1^{(2)}(\gamma\Pi + m)^{-1} \\ &= \frac{\alpha}{4\pi} \int_0^1 du \left[\frac{4}{u} + \frac{(\gamma\Pi)^2 u + m\gamma\Pi(2 + u) - 2m^2}{m^2 - (1 - u)(\gamma\Pi)^2} \right]. \end{aligned} \quad (\text{A.8})$$

The first term of the integral is infrared divergent. But it is independent of Π so that

$$\delta_A[M_1^{(2)}G_0] = \delta_A \frac{\alpha}{4\pi} \int_0^1 du \frac{(\gamma\Pi)^2 u + m\gamma\Pi(2+u) - 2m^2}{m^2 - (1-u)(\gamma\Pi)^2} \quad (\text{A.9})$$

which is finite. We may take the derivative and insert the resulting expression into (5). When δ_A acts on Π it replaces it with $e^{-ikx}\gamma_\mu$. The displacement property of this operator implies that each Π standing between the two γ_μ 's will be shifted to $\Pi - k$. After the k -integration is done we set $\gamma\Pi = -m$. The term $G_0^{-1} = \gamma\Pi + m$, on the far right of (5), is not affected by the k -shift and integration so that it will cause the whole expression to vanish unless the integral standing to the left of it has a pole at $\gamma\Pi = -m$. In the present case the k -integral turns out to be quite finite at $\gamma\Pi = -m$ so there is no contribution at all to the magnetic moment.

We next consider $M_F^{(2)}$ as given by Eq. (3.24):

$$M_F^{(2)} = -\frac{\alpha}{4\pi} \int_0^1 du (1-u) \frac{4m\mathfrak{F}' + (2-u)\{\mathfrak{F}', \gamma\Pi\}}{m^2 - (1-u)(\gamma\Pi)^2}. \quad (\text{A.10})$$

The prime on $\mathfrak{F}'$ signifies that we are to treat it as being unaffected by the operation δ_A . Then $M_F^{(2)}$ commutes with G_0 and we may write

$$\begin{aligned} \{\delta_A[M_F^{(2)}G_0]\}G_0^{-1} &= \{\delta_A[G_0M_F^{(2)}]\}G_0^{-1} \\ &= G_0\gamma_\mu e^{-ikx}M_F^{(2)} + G_0[\delta_A M_F^{(2)}]G_0^{-1}. \end{aligned} \quad (\text{A.11})$$

Placing the quantity $G_0\gamma_\mu e^{-ikx}M_F^{(2)}$ in (5) we obtain

$$\begin{aligned} -i(\alpha/4\pi^3) \int (dk)(k^2)^{-1} e^{ikx} \gamma_\mu G_0 G_0 \gamma_\mu e^{-ikx} M_F^{(2)} \\ = -i(\alpha/4\pi^3) \int (dk)(k^2)^{-1} \gamma_\mu [\gamma(\Pi - k) + m]^{-2} \gamma_\mu \quad (\text{A.12}) \\ = [(\partial/\partial m)M_0^{(2)}]M_F^{(2)}. \end{aligned}$$

When we set $\gamma\Pi = -m$ this will just cancel the contribution to the self-energy from M_M as given in (2.40), if we use the definition (2.15) of $\tilde{M}_0^{(2)}$.

The other term, $G_0[\delta_A M_F^{(2)}]G_0^{-1}$ will produce no contribution to the moment because of the factor $\gamma\Pi + m$ on the right. In this case, it turns out that the k -integral behaves like $\log[1 - (\gamma\Pi)^2/m^2]$ as $\gamma\Pi \rightarrow -m$, but this is not sufficiently singular to counteract the zero of $(\gamma\Pi + m)$.

After all these terms have been disposed of, what remains is simply

$$M_G + M_\Gamma + M_M = -i(\alpha/4\pi^3) \int (dk)(k^2)^{-1} \gamma_\mu e^{ikx} G_0 [\delta_A \tilde{M}^{(2)}]. \quad (\text{A.13})$$

As we have said, it is extremely difficult, if not impossible to calculate $\tilde{M}^{(2)}$ directly. But we can find $\delta_A \tilde{M}^{(2)}$. We shall indicate how this may be done. The second-order mass operator is

$$M^{(2)} = -i(\alpha/4\pi^3) \int_0^\infty ds \int_0^1 du \int (dk') \gamma_\nu [m - \gamma(\Pi - k')] \quad (\text{A.14})$$

$$\times \exp \{ -isu[m^2 + (\Pi - k')^2 - \mathfrak{F}] \} \gamma_\nu \exp(-ik'^2).$$

We may not use the fact that $\mathfrak{F}$ commutes with Π until after δ_A has been applied. Instead, following Newton (9), we write more generally, correct to second degree in F

$$\exp \{ -isu[(\Pi - k')^2 - \mathfrak{F}] \} = \exp [-isu(\Pi - k')^2]$$

$$+ isu \int_0^1 dv \exp [-isu(1 - v)(\Pi - k')^2] \mathfrak{F} \exp [-isuv(\Pi - k')^2]$$

$$+ (isu)^2 \int_0^1 dv \exp [-isu(1 - v)(\Pi - k')^2] \mathfrak{F} \int_0^v dw \quad (\text{A.15})$$

$$\times \exp [-isuv(v - w)(\Pi - k')^2] \mathfrak{F} \exp [-isuw(\Pi - k')^2].$$

Using the fact that $\delta_A(\mathfrak{F}) = e^{-ikx} i\sigma_{\mu\lambda} k_\lambda$, we obtain

$$\delta_A M^{(2)} = -i(\alpha/4\pi^3) \delta_A \int_0^\infty s ds \int_0^1 du \int (dk') \exp [-isum^2]$$

$$\times \gamma_\nu [m - \gamma(\Pi - k')] (1 + isu\mathfrak{F}') \gamma_\nu \exp \{ -is[(k' - u\Pi)^2$$

$$+ u(1 - u)\Pi^2] \} + K_1, \quad (\text{A.16})$$

where

$$K_1 = i(\alpha/4\pi^3) e^{-ikx} \int_0^\infty s^2 ds \int_0^1 u du \int_0^1 dv \int (dk') \gamma_\nu [m - \gamma(\Pi - k - k')] \quad (\text{A.17})$$

$$\times [\sigma_{\mu\lambda} k_\lambda + isuv\sigma_{\mu\lambda} k_\lambda \mathfrak{F} + isu(1 - v)\mathfrak{F}\sigma_{\mu\lambda} k_\lambda] \gamma_\nu$$

$$\times \exp [-isu(1 - v)(\Pi - k - k')^2] \exp [-isuv(\Pi - k')^2].$$

If we can extract the terms $\delta_A[M_0^{(2)} + M_F^{(2)}]$ explicitly from $\delta_A[M^{(2)}]$, then we will be left with $\delta_A[\tilde{M}^{(2)}]$. To this end we note that the ν -summation in (A.16) is the same as that which occurred in (3.13). If we write

$$-4m - 2\gamma(\Pi - k') + 2isu\mathfrak{F}'\gamma(\Pi - k')$$

$$= -4m - 2\gamma\Pi(1 - u) + 2isu(1 - u)\mathfrak{F}'\gamma\Pi + 2(1 - isu\mathfrak{F}')\gamma(k' - u\Pi),$$

and, following Karplus, Klein, and Schwinger (10),

$$\begin{aligned} & \exp \{ -is[(k' - u\Pi)^2 + u(1 - u)\Pi^2] \} \\ &= \exp \{ -is[k'^2 + u(1 - u)\Pi^2] \} + \int_0^1 dy (\partial/\partial y) \\ & \quad \times \exp \{ -is[(k' - uy\Pi)^2 + u(1 - u)\Pi^2] \}, \end{aligned} \quad (\text{A.18})$$

then we may extract from $\delta_A M^{(2)}$ the following part:

$$\begin{aligned} \delta_A M^{(2)} - K_1 - K_2 - K_3 &= -i(\alpha/4\pi^3)\delta_A \int_0^\infty s ds \int_0^1 du \int (dk') \\ & \quad \times \exp(-isk'^2 - isum^2)[-4m - 2\gamma\Pi(1 - u) \\ & \quad + 2isu(1 - u)\mathfrak{F}'\gamma\Pi] \exp[isu(1 - u)\Pi^2] = -(\alpha/4\pi)\delta_A \int_0^\infty ds s^{-1} \\ & \quad \times \int_0^1 du \exp(-isum^2)[-4m - 2\gamma\Pi(1 - u) \\ & \quad + 2isu(1 - u)\mathfrak{F}'\gamma\Pi] \exp[-isu(1 - u)\Pi^2], \end{aligned} \quad (\text{A.19})$$

where

$$\begin{aligned} K_2 &= -i(\alpha/4\pi^3)\delta_A \int_0^\infty s ds \int_0^1 du \int (dk') \exp(-isum^2) \\ & \quad \times 2(1 - isu\mathfrak{F}')\gamma(k' - u\Pi)e^{-is\phi(1)} \end{aligned} \quad (\text{A.20})$$

and

$$\begin{aligned} K_3 &= -i(\alpha/4\pi^3)\delta_A \int_0^\infty s ds \int_0^1 du \int_0^1 dy \int_0^1 dw \int (dk') \exp(-isum^2) \\ & \quad \times [-4m - 2\gamma\Pi(1 - u) + 2isu(1 - u)\mathfrak{F}'\gamma\Pi](-is) \\ & \quad e^{-is(1-u)\phi(y)}[(\partial/\partial y)\phi(y)]e^{-isw\phi(y)}. \end{aligned} \quad (\text{A.21})$$

We have used the abbreviation

$$\phi(y) = (k - uy\Pi)^2 + u(1 - u)\Pi^2 \quad (\text{A.22})$$

and the formula

$$(\partial/\partial y)e^{-is\phi(y)} = -is \int_0^1 dw e^{-is(1-w)\phi(y)}[\partial\phi(y)/\partial y]e^{-isw\phi(y)}. \quad (\text{A.23})$$

We may write

$$\begin{aligned}
\exp[-isu(1-u)\Pi^2] &= \exp\{-isu(1-u)[\mathfrak{F} - (\gamma\Pi)^2]\} \\
&= \exp[isu(1-u)(\gamma\Pi)^2] - isu(1-u) \int_0^1 dv \\
&\quad \exp[isu(1-u)(1-v)(\gamma\Pi)^2] \mathfrak{F} \exp[isu(1-u)v(\gamma\Pi)^2] \\
&\quad + [-isu(1-u)]^2 \int_0^1 dv \int_0^v dw \exp[is(1-v)u(1-u)(\gamma\Pi)^2] \mathfrak{F} \\
&\quad \times \exp[is(v-w)u(1-u)(\gamma\Pi)^2] \mathfrak{F} \exp[iswu(1-u)(\gamma\Pi)^2]
\end{aligned} \tag{A.24}$$

in order to replace Π^2 by $(\gamma\Pi)^2$. Using this in (19) and taking the derivative of $\mathfrak{F}$ explicitly, we obtain

$$\begin{aligned}
\delta_A M^{(2)} - K_1 - K_2 - K_3 - K_4 &= -(\alpha/4\pi)\delta_A \int_0^\infty dss^{-1} \int_0^1 du \\
&\quad \exp(-ism^2)[-4m - 2\gamma\Pi(1-u) + 4ismu(1-u)\mathfrak{F}' \\
&\quad + 2isu(1-u)^2\gamma\Pi\mathfrak{F}' + 2isu(1-u)\mathfrak{F}'\gamma\Pi] \exp[isu(1-u)(\gamma\Pi)^2]
\end{aligned} \tag{A.25}$$

where

$$\begin{aligned}
K_4 &= -(\alpha/4\pi)e^{-ikx} \int_0^\infty ds \int_0^1 du(1-u)u \int_0^1 dv \exp(-ism^2) \\
&\quad \times [-4m - 2\gamma(\Pi - k)(1-u) + 2isu(1-u)\mathfrak{F}'\gamma(\Pi - k)] \\
&\quad \times \exp\{is(1-v)u(1-u)[\gamma(\Pi - k)]^2\} \\
&\quad \times [\sigma_{\mu\lambda}k_\lambda - isu(1-u)v\sigma_{\mu\lambda}b_\lambda\mathfrak{F}' - isu(1-u)(1-v)\mathfrak{F}'\sigma_{\mu\lambda}k_\lambda] \\
&\quad \times \exp[isvu(1-u)(\gamma\Pi)^2].
\end{aligned} \tag{A.26}$$

Finally, we write

$$\begin{aligned}
\delta_A M^{(2)} - K_1 - K_2 - K_3 - K_4 - K_5 &= -(\alpha/4\pi)\delta_A \int_0^\infty dss^{-1} \int_0^1 du \exp(-ism^2) \\
&\quad \times [-4m - 2\gamma\Pi(1-u) + 4isu(1-u)\mathfrak{F}' + isu(1-u) \\
&\quad \times (2-u)\{\mathfrak{F}', \gamma\Pi\}] \exp[isu(1-u)(\gamma\Pi)^2],
\end{aligned} \tag{A.27}$$

$$\begin{aligned}
K_5 &= -i(\alpha/4\pi)\delta_A \int_0^\infty ds \int_0^1 duu^2(1-u) \exp(-ism^2)[\mathfrak{F}', \gamma\Pi] \\
&\quad \times \exp[isu(1-u)(\gamma\Pi)^2].
\end{aligned} \tag{A.28}$$

We recognize that after the subtractions of the first two powers of $(\gamma\Pi + m)$ from the field-independent part of (27), the quantity remaining to be differentiated is precisely $M_1^{(2)} + M_F^{(2)}$ whose effects we have already taken into account. Thus we may conclude

$$\delta_A \tilde{M}^{(2)} = K_1 + K_2 + K_3 + K_4 + K_5. \quad (\text{A.29})$$

All that remains is to substitute this into (13) and carry out the integrations.

As a concluding remark we present a formula to be used in differentiating the exponentials of (20), (21), and (28):

$$\delta_A e^x = \int_0^1 dv e^{(1-v)x} \delta_A \chi e^{vx}. \quad (\text{A.30})$$

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